

David Williams

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SUMMARY

Senior Data Engineer with **10+ years of experience building large-scale distributed data platforms and high-performance data pipelines** across fintech and healthcare industries. Expert in designing **cloud-native lakehouse architectures, real-time streaming systems, and large-scale ETL/ELT frameworks** that power analytics, machine learning, and operational intelligence. Proven track record delivering **petabyte-scale data infrastructure** using **Apache Spark, Kafka, Airflow, and Databricks** across AWS, Azure, and GCP. Specialized in **distributed data processing, data platform architecture, streaming systems, and data reliability frameworks**, enabling organizations to process **multi-terabyte datasets daily while improving performance, reliability, and cost efficiency**.

TECHNICAL SKILLS

- **Programming Languages:** Python, Scala, SQL, Java
- **Data Processing & Big Data:** Apache Spark, Spark Structured Streaming, Databricks, Delta Lake, Distributed Data Processing
- **Streaming & Event Architecture:** Apache Kafka, Event-Driven Architecture, Real-Time Data Pipelines
- **Workflow Orchestration & Data Transformation:** Apache Airflow (MWAA), dbt, DAG Design, ELT/ETL Frameworks
- **Cloud Platforms:** AWS (S3, EMR, Glue, Lambda, Redshift, MWAA), Microsoft Azure, Google Cloud Platform (GCP)
- **Data Warehousing & Storage:** Snowflake, Amazon Redshift, BigQuery, Data Lakes, Lakehouse Architecture, Dimensional Modeling
- **Infrastructure & DevOps:** Docker, Kubernetes, Terraform, Infrastructure as Code, CI/CD
- **Observability & Data Reliability:** Data Quality Frameworks, Schema Enforcement, Data Validation, Monitoring, Prometheus, Grafana

WORK EXPERIENCE

Plaid

Jul 2024 – Present

Senior Data Engineer

Remote

- Architect and operate an **AWS-native financial data platform processing 6+ TB of transaction data daily** powering analytics, fraud detection, and operational intelligence.
- Design and maintain **150+ production-grade Airflow pipelines** supporting batch and near real-time data processing workloads.
- Build scalable **data lake and warehouse architecture using Amazon S3 and Redshift**, enabling enterprise analytics and BI platforms.
- Implement **event-driven ingestion pipelines using Apache Kafka**, enabling near real-time financial data processing.
- Optimize distributed **Spark workloads and partitioning strategies**, reducing infrastructure costs by **30%** while improving runtime performance.
- Develop **curated data models and analytical datasets** used by product, analytics, and machine learning teams.
- Implement **data validation, monitoring, and alerting frameworks**, reducing pipeline failures by **45%**.
- Collaborate cross-functionally with product, analytics, and ML teams to design **scalable data services and data contracts**.

Komodo Health

Jan 2021 – Jul 2024

Big Data Engineer

Remote

- Designed and built distributed **Spark data pipelines processing 20+ TB healthcare datasets**, supporting large-scale analytics and machine learning workloads.
- Architected enterprise **AWS data lake platform (S3, EMR, Glue)** serving **500+ healthcare enterprise customers**.

- Led migration of legacy ETL pipelines to **Databricks Lakehouse architecture**, improving pipeline performance by **60%**.
- Built **real-time streaming pipelines using Kafka and Spark Structured Streaming** for near real-time healthcare data ingestion.
- Developed complex **Airflow DAGs orchestrating batch and streaming workflows** with SLA monitoring, retries, and automated backfills.
- Implemented **data quality and validation frameworks**, reducing production data incidents by **50%**.
- Created **reusable pipeline templates and engineering standards**, improving team productivity and deployment speed.
- Partnered with data scientists and analytics teams to deliver **scalable data infrastructure for advanced healthcare insights**.

Databricks

May 2017 – Dec 2020

Data Platform Engineer

San Francisco, CA

- Architected and operated **petabyte-scale cloud data platform infrastructure** supporting distributed analytics workloads.
- Built **Spark-based batch and streaming pipelines** ingesting large-scale datasets from APIs, operational databases, and file-based sources.
- Developed internal **data platform frameworks and tooling** enabling **20+ engineering teams** to deploy scalable data workflows.
- Integrated cloud warehouses including **Snowflake and Amazon Redshift** to power enterprise analytics platforms.
- Implemented **data reliability frameworks** including schema enforcement, deduplication, idempotent processing, and automated quality checks.
- Led **Spark performance optimization initiatives**, reducing cluster runtime by **40%**.
- Built centralized **monitoring, logging, and operational dashboards**, improving platform observability and reliability.
- Mentored engineers on **distributed systems design and Spark optimization best practices**.

Health Catalyst

Oct 2015 – May 2017

Analytics Engineer

South Jordan, Utah

- Developed **SQL and Python ETL pipelines** ingesting multi-terabyte healthcare datasets into enterprise data warehouse systems.
- Designed **dimensional data models** supporting business intelligence and operational analytics.
- Optimized distributed transformation workflows, improving pipeline performance by **65%**.
- Implemented **automated data validation frameworks** to improve reliability of analytics datasets.
- Partnered with analytics teams to deliver **trusted enterprise data models and reporting datasets**.

EDUCATION

University of Texas – Arlington

Bachelor of Science, Computer Science

2011 - 2015